

Power Electronics - Lab

Open Ended Lab Report



Designing a Variable Voltage Regulator

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Introduction

This project aims to design and implement a discrete component linear voltage regulator capable of supplying a stable 5.0 V output from a 10.0 V DC input. The design utilizes basic components like BJTs, operational amplifiers, and Zener diodes to mimic the function of commercial voltage regulators, with added emphasis on performance analysis such as load and line regulation, and short-circuit protection.

Problem Analysis

A regulated 5.0 V supply is required in many embedded and analog systems. The challenge is to build such a supply using only discrete components while ensuring voltage stability despite changes in load and supply. Furthermore, the design must be protected against short circuits to enhance reliability.

Design Requirements

- Input Voltage: 10.0 V DC
- Output Voltage: 5.0 V DC
- Components: Discrete (BJTs, Op-Amps, Zener Diodes, Resistors)
- Features: Load regulation, line regulation, short-circuit protection

Feasibility Analysis

- **Time:** The design was completed within the lab session.
- **Cost:** All components are low-cost and readily available.
- **Skills:** Requires basic understanding of op-amp circuits, BJTs, and feedback control.

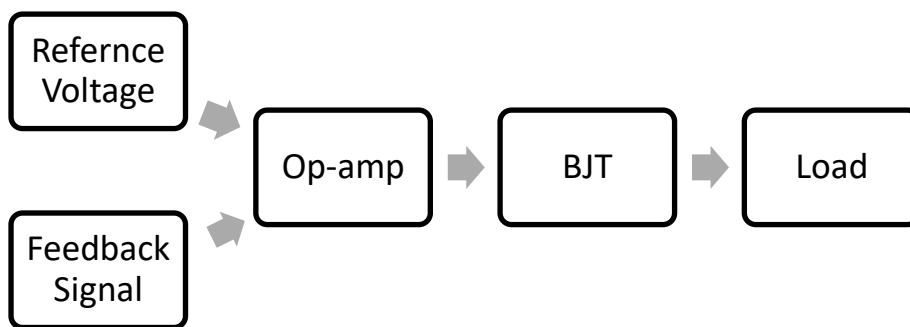
Possible Solutions

Several approaches were considered:

- Using only BJTs (difficult to stabilize output)
- Using an op-amp feedback mechanism with a pass transistor.

The second approach was chosen due to better voltage regulation performance and ease of integrating protection mechanisms.

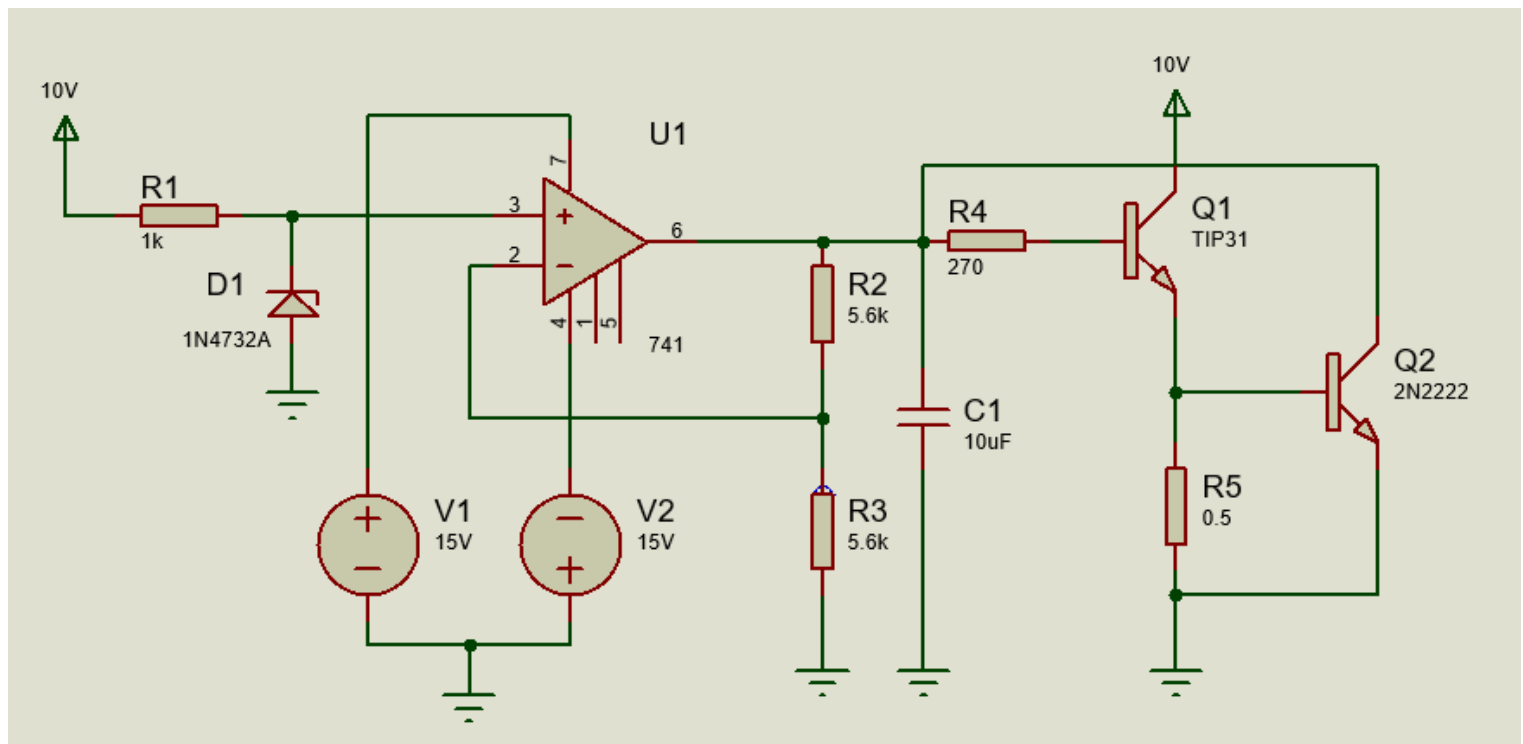
Preliminary Design



Design Description

The op-amp compares the output voltage (via resistor divider) to the 5.1 V zener reference. It drives the base of a power NPN transistor, regulating the output. A low-value resistor in series with the load senses current; when excessive current causes >0.6 V drop, a secondary transistor shunts the base of the main BJT to ground, limiting current.

Circuit Design



Performance Analysis

- **Load Regulation:** Excellent; $< 1\%$ variation
- **Line Regulation:** Good; output remains within tolerance
- **Short-Circuit Protection:** Limits current effectively without damage

Conclusion

A discrete 5.0 V linear voltage regulator was successfully designed and tested. It performed well under varying line and load conditions and included a simple yet effective short-circuit protection mechanism.

